BetterCode: Reproducible

Someone else, or future you, can rerun everything.

Overview

- *Recap day 1+2* - readable, reusable
  - Help with code comprehension & abstraction
  - Logical workflow
- **Reproducible**: Environment
  - Help with execution
  - Activity 1: Build a conda environment
  - Activity 2: Turn code to command-line tool

Reproducible issue in (computational) research

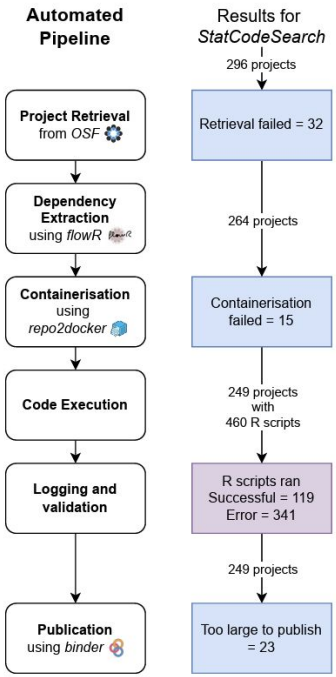


Figure 1: Our completely automated pipeline `osf-to-binder` (left) and the analysis results using this pipeline for the *StatCodeSearch* dataset (right).

Saju, Lorraine, et al. "Computational Reproducibility of R Code Supplements on OSF." arXiv preprint arXiv:2505.21590 (2025).

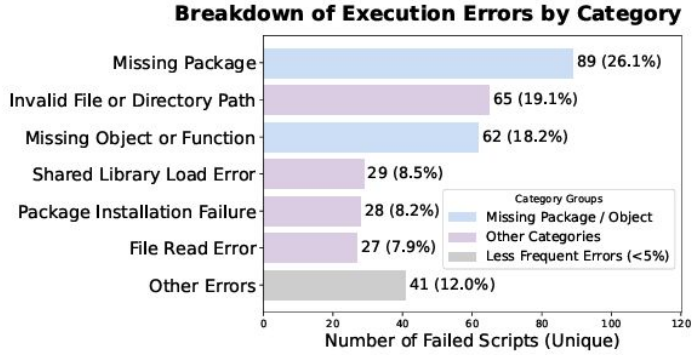
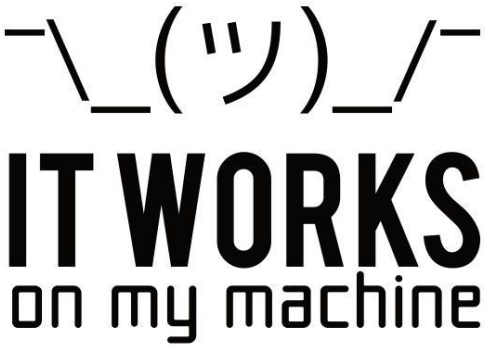


Figure 2: Breakdown of execution errors by category.

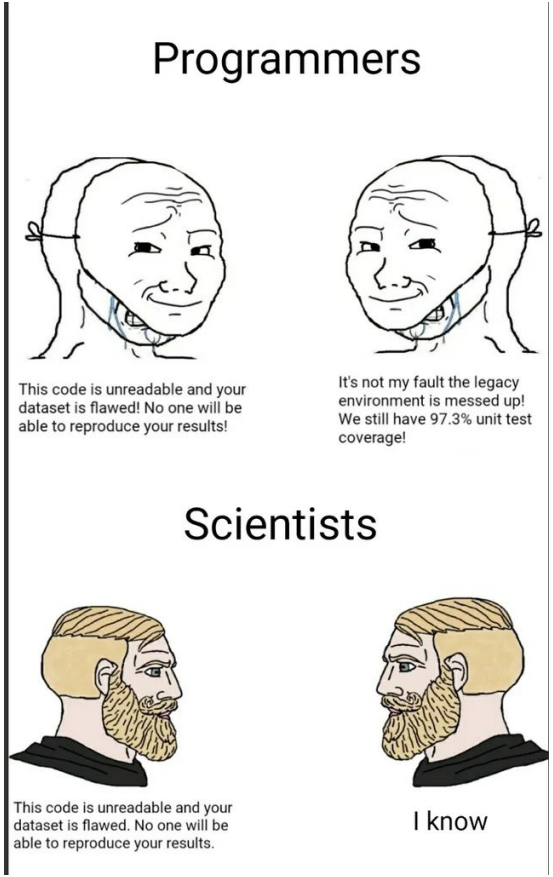
Why is reproducible important?

- **Foundation** of good science & engineering
- Trust & credibility
- Debugging becomes possible
- Collaboration works
- Saves time, less rework



What Reproducibility Needs?

- **Defined environment**
- **Executable code that run end-to-end**
- Predictable behaviors
- Proper documentation



What happens when a code is executed?

Layer	What it defies	Python example	R example
Code	Intent	*pd.read_csv("data.csv")*	*read_csv("data.csv")*
Package/Library	Algorithms + implementation	*pandas::read_csv()*	*readr::read_csv()*
Environment	Package versions	*pandas == 1.5*	*readr 2.1.4*
System libraries	Performance + computation	*libc* (file I/O)	
Operating system	Resources	*Linux* file system	
Hardware	Physical execution		

Popular options to manage your environment

Layer	What it defies	Python	R
Code	Intent	(your code)	(your code)
Package/Library	Algorithms + implementation	PyPI	CRAN
Environment	Package versions	venv, **uv** …	**renv**
System libraries	Performance + computation	**conda**, mamba, pixi, …	
Operating system	Resources	containers (**singularity**, Docker)	
Hardware	Physical execution	Physical machine, cloud hardware	

Singularity / Apptainer

- How to build a container: resources on MyJAX

https://jacksonlaboratory.sharepoint.com/sites/ResearchIT/SitePages/Containerization-with-Singularity--101.aspx

- Truly reproducible!
- Portable
- Slow iteration, not flexible
- Large image files

Conda

- How to: bestpractices_workshop

https://github.com/TheJacksonLaboratory/bestpractices_workshop/blob/main/conda.md

- Flexible
- Mixed-language workflows
- Large disk usage
- Slow dependency resolution

Language-specific package management

- Quick & (should be) easy
- Project-level isolation
- Language specific
- uv (and other Python-based venv) provides more enviromental control

My recommendation

In development work:

- conda
- Python-only: uv or other venv management
- R-only: conda + renv

“Established” pipeline: container

Demo 1: Creating environment

Three ways of making setup more explicit:

- `renv` for an R project
- `uv` for a Python project
- `conda` environment

What should I commit, and what should another person recreate locally?

Demo 1 - Tool comparisons

Tool	Shareable requirements	Exact resolved version	Local setup
renv	renv.lock	renv.lock	renv/library
uv	uv.lock & pyproject.toml	uv.lock	.venv/
conda	environment.yml		[path to your conda]

Activity 1 - Create a Conda Environment

1. Create a conda environment and provide the yaml file.

2. Recreate an environment using a yaml file

Demo 2: Script entrypoint

We have developed and rerun the same analysis pipeline with different ways

- Open an editor, run codes
- Parameterized Notebooks

We now want to make it into a “tool”

- Rscript analysis.R --input data.csv --output results.pdf
- Python analysis.py --input data.csv --output results.pdf

Activity 2 - Write an Executable Script

You have a working code to create plots.

Turn it into an executable command-line tool

Take-away message

- Good reproducibility makes setup, inputs, outputs, and run order visible